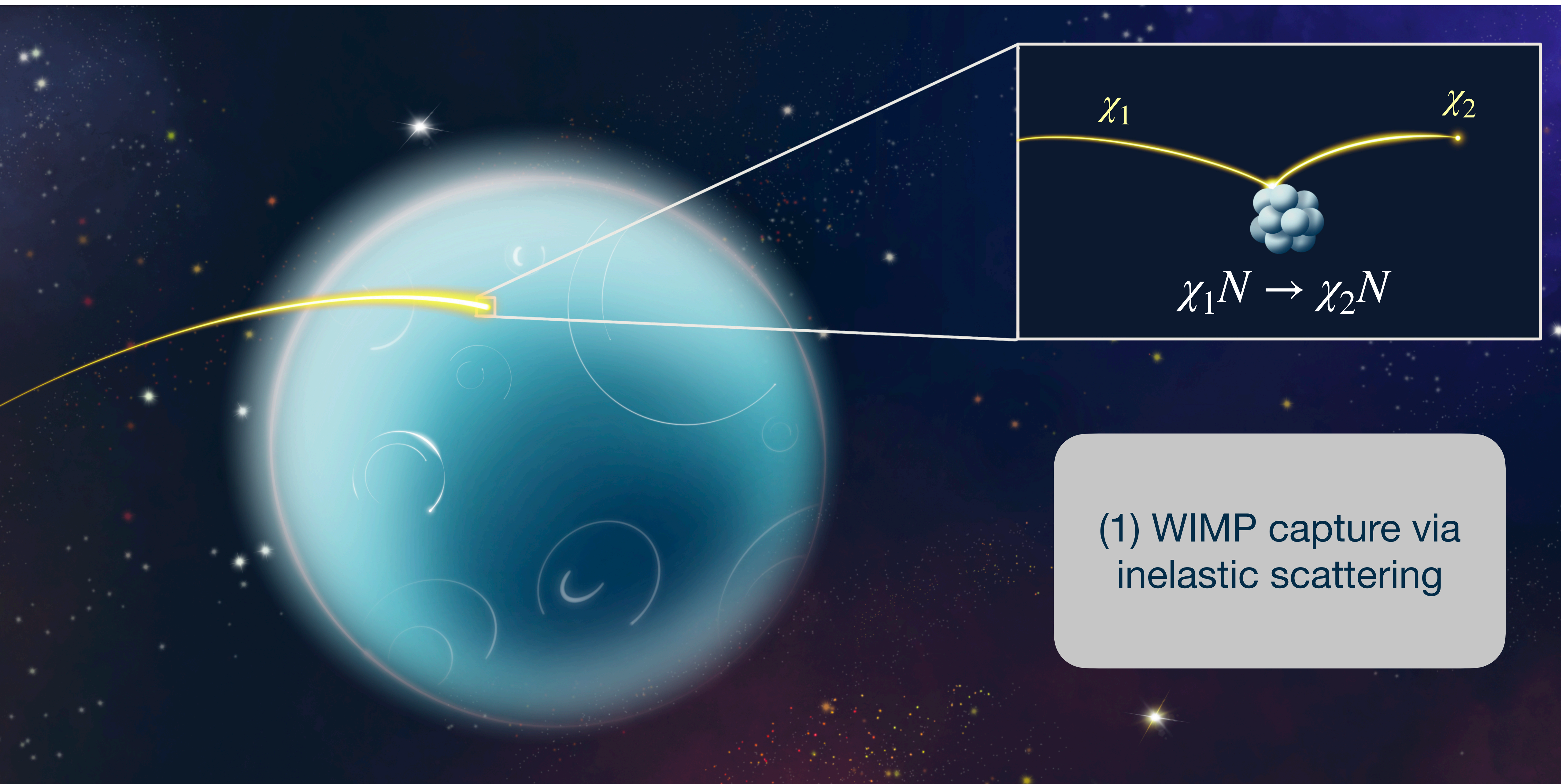


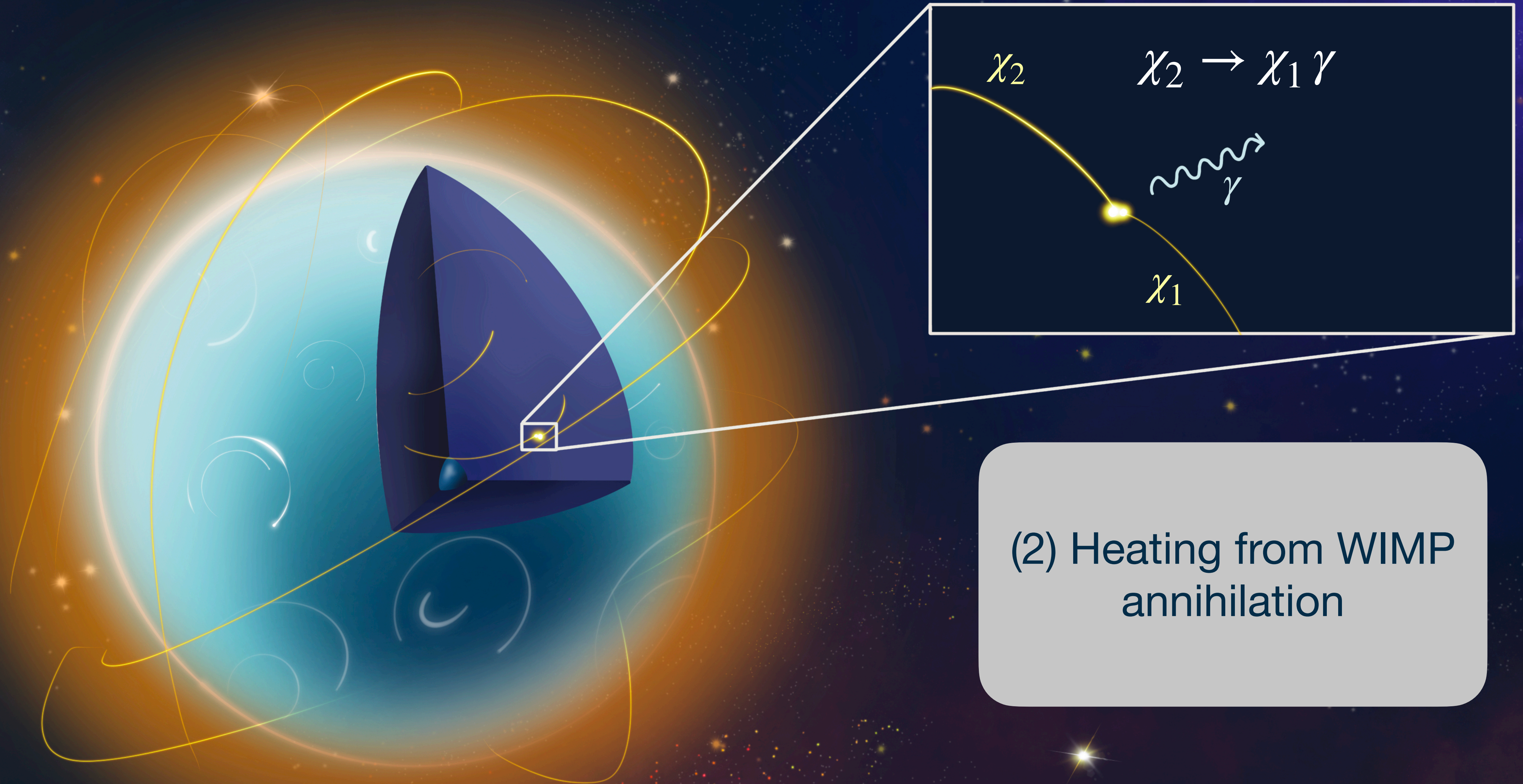


Detecting WIMPs with White Dwarf heating in ω Cen

Motivation: White Dwarf heating from WIMPs



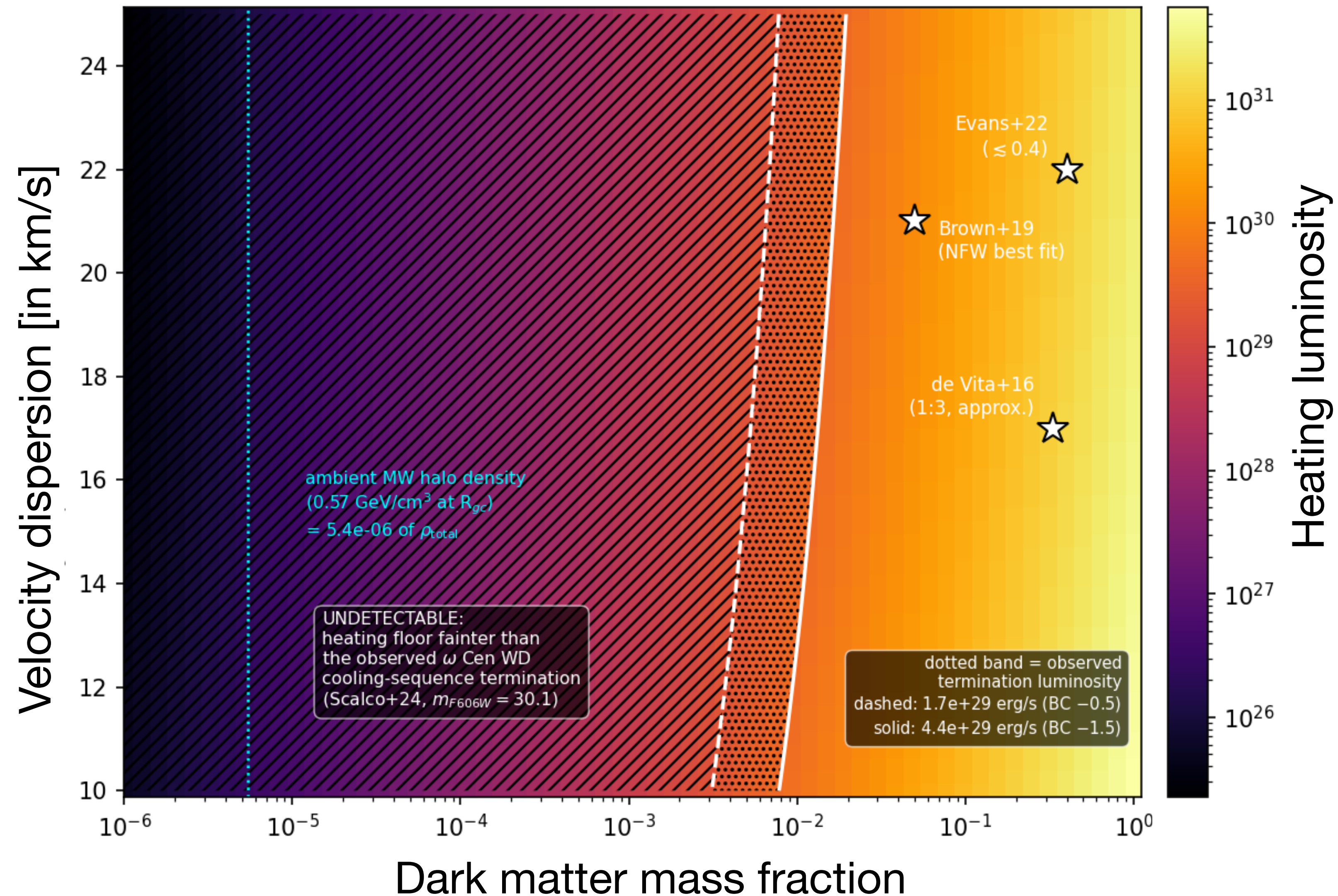
Motivation: White Dwarf heating from WIMPs



Target: White Dwarfs in ω Cen

E.g.: $\sigma = 10^{-38} \text{ cm}^2$, $\delta = 200 \text{ keV}$, $m_\chi = 1.1 \text{ TeV}$

- ω Cen would be a great target (vicinity to us, high stellar mass so lots of WDs) *e.g. Krall & Reece, 2018*
- Cooling sequence has been observed *e.g. Scalco et al. 2024*



Dark (matter?) component of ω Cen

Intermediate mass black hole?

Article | [Open access](#) | Published: 10 July 2024

Fast-moving stars around an intermediate-mass black hole in ω Centauri

[Maximilian Häberle](#) , [Nadine Neumayer](#), [Anil Seth](#), [Andrea Bellini](#), [Mattia Libralato](#), [Holger Baumgardt](#), [Matthew Whitaker](#), [Antoine Dumont](#), [Mayte Alfaro-Cuello](#), [Jay Anderson](#), [Callie Clontz](#), [Nikolay Kacharov](#), [Sebastian Kamann](#), [Anja Feldmeier-Krause](#), [Antonino Milone](#), [Maria Selina Nitschai](#), [Renuka Pechetti](#) & [Glenn van de Ven](#)

New constraints on the central mass contents of Omega Centauri from combined stellar kinematics and pulsar timing

 Andrés Bañares-Hernández^{1,2★},  Francesca Calore³, Jorge Martin Camalich^{1,2,4} and Justin I. Read⁵

Gemini and *Hubble Space Telescope* Evidence for an Intermediate-Mass Black Hole in ω Centauri

Eva Noyola, Karl Gebhardt, and Marcel Bergmann

ω Centauri: a MUSE discovery of a counter-rotating core

[Renuka Pechetti](#) , [Sebastian Kamann](#) , [Davor Krajnović](#) , [Anil Seth](#), [Glenn van de Ven](#), [Nadine Neumayer](#), [Stefan Dreizler](#), [Peter M Weilbacher](#), [Sven Martens](#), [Florence Wragg](#)

Dark (matter?) component of ω Cen

Dark matter annihilation?

The glow of annihilating dark matter in Omega Centauri

[Anthony M. Brown](#), [Richard Massey](#), [Thomas Lacroix](#), [Louis E. Strigari](#), [Azadeh Fattahi](#), [Céline Boehm](#)

On the origin of the gamma-ray emission from Omega Centauri:
millisecond pulsars and dark matter annihilation

Javier Reynoso-Cordova, Oleg Burgueño, Alex Geringer-Sameth, Alma X. González-Morales,
Stefano Profumo and O. Valenzuela

Discovery of Millisecond Pulsars in the Globular Cluster Omega Centauri

[Shi Dai](#), [Simon Johnston](#), [Matthew Kerr](#), [Fernando Camilo](#), [Andrew Cameron](#), [Lawrence Toomey](#), [Hiroki Kumamoto](#)

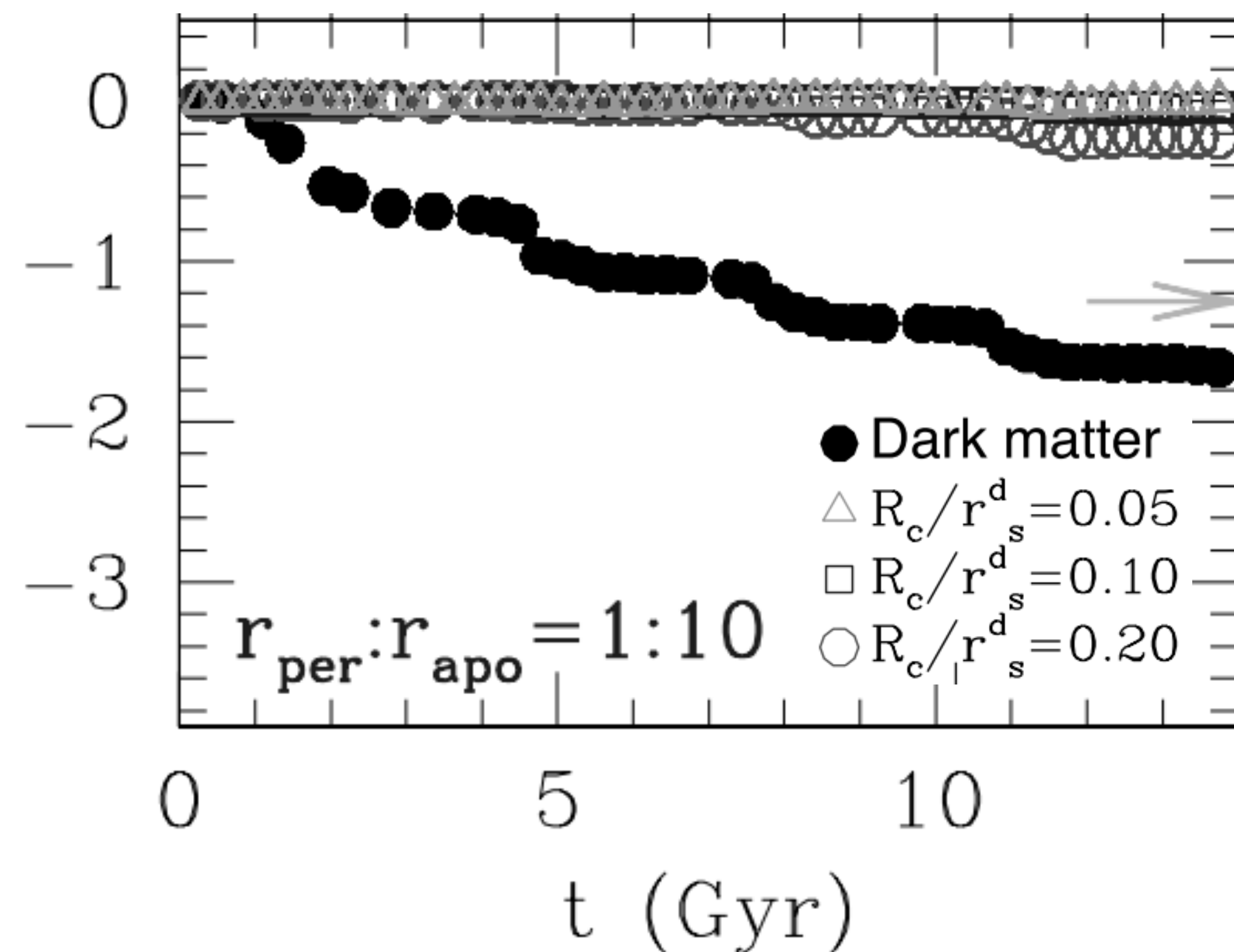
Constraints on annihilating dark matter in the Omega Centauri cluster

[Man Ho Chan](#)  and [Chak Man Lee](#)

Idea: Ab initio estimate of DM content from tidal stripping

Peñarrubia, Navarro & McConnachie, 2007

The tidal evolution of local group dwarf spheroidal

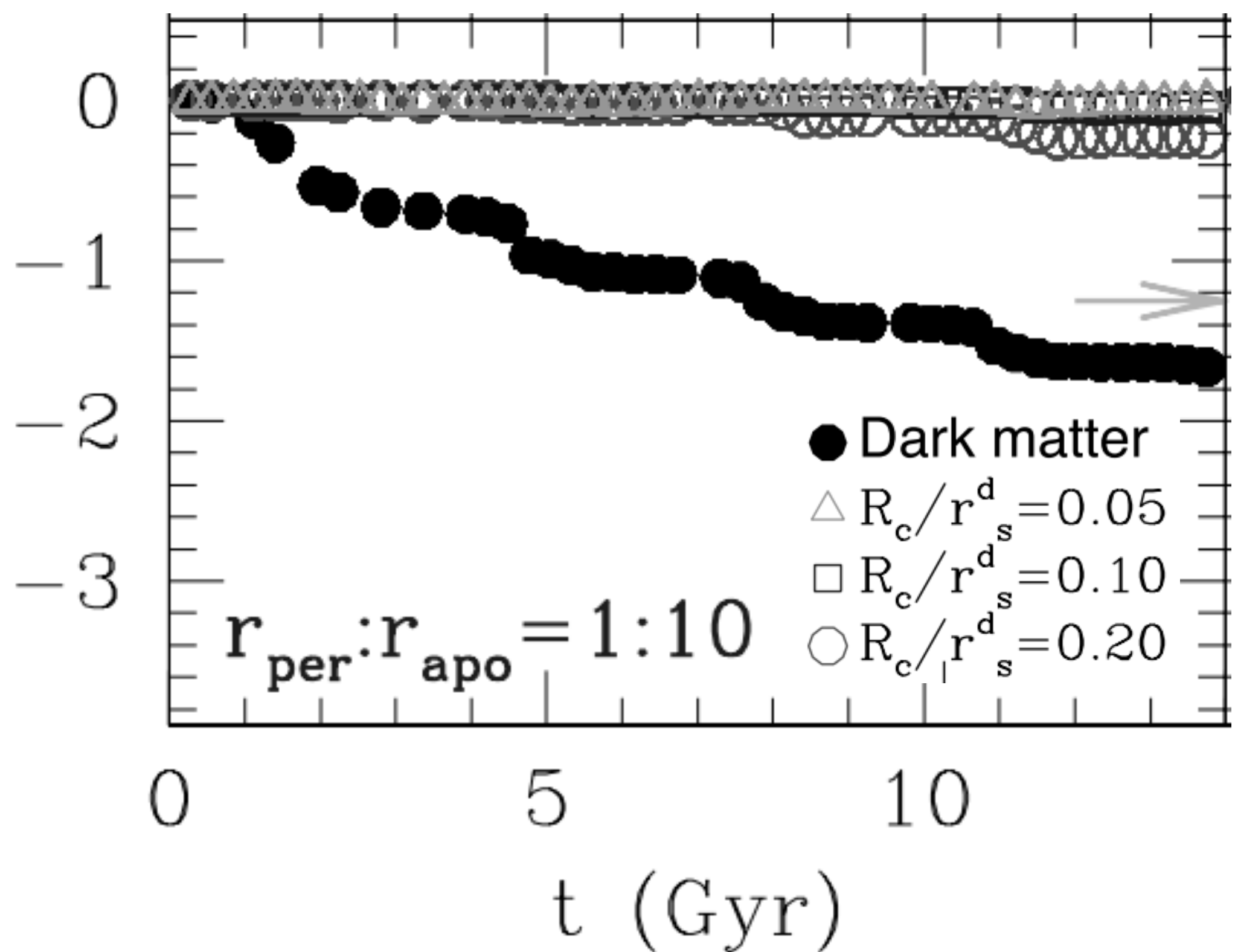


Idea: Ab initio estimate of DM content from tidal stripping

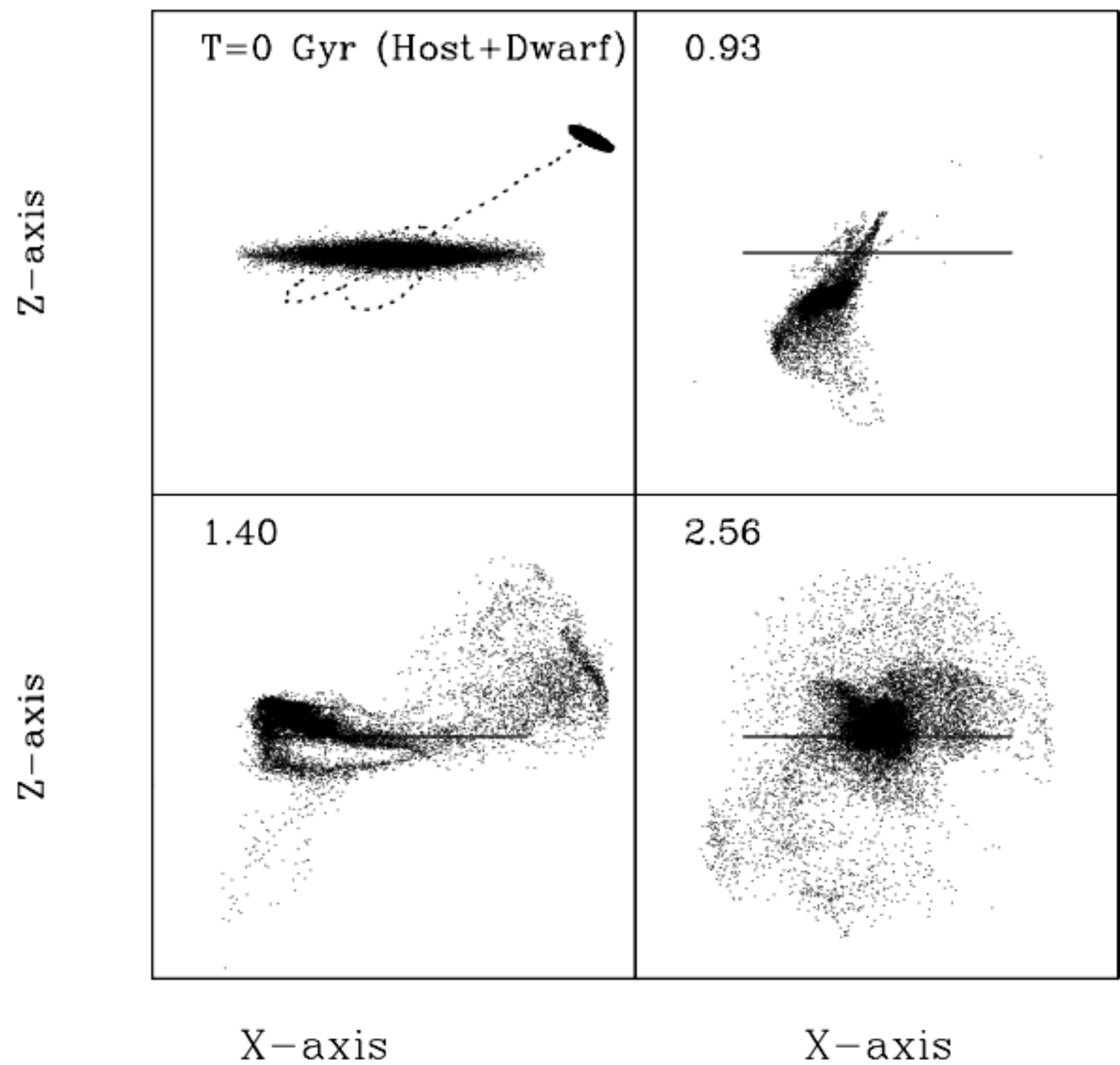
Bekki & Freeman, 2003

Peñarrubia, Navarro & McConnachie, 2007

The tidal evolution of local group dwarf spheroidals



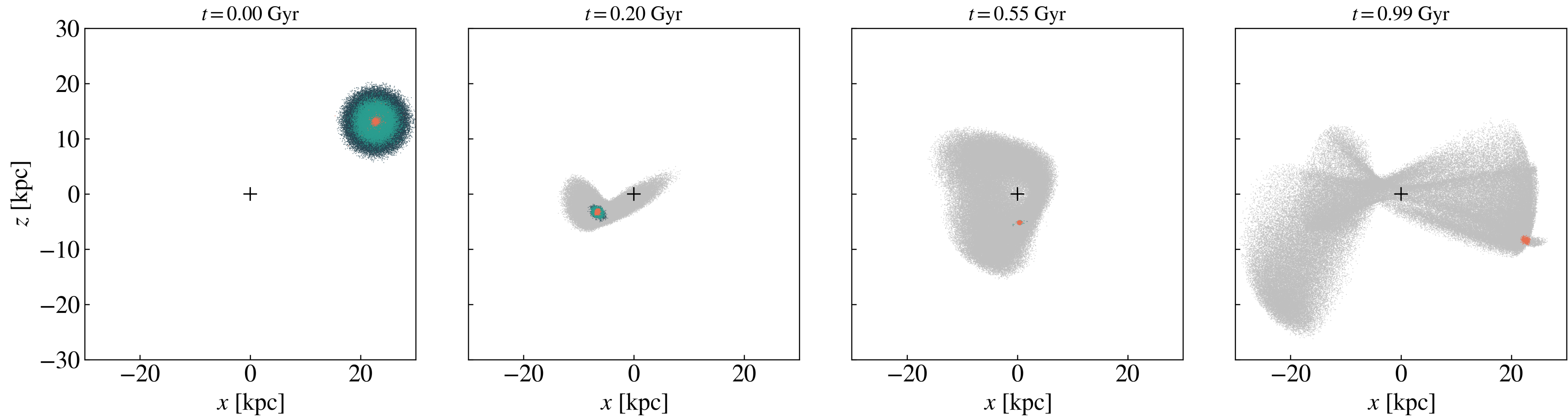
Formation of ω Cen from an ancient nucleated dwarf galaxy in the young galactic disc



Idea: Ab initio estimate of DM content from tidal stripping

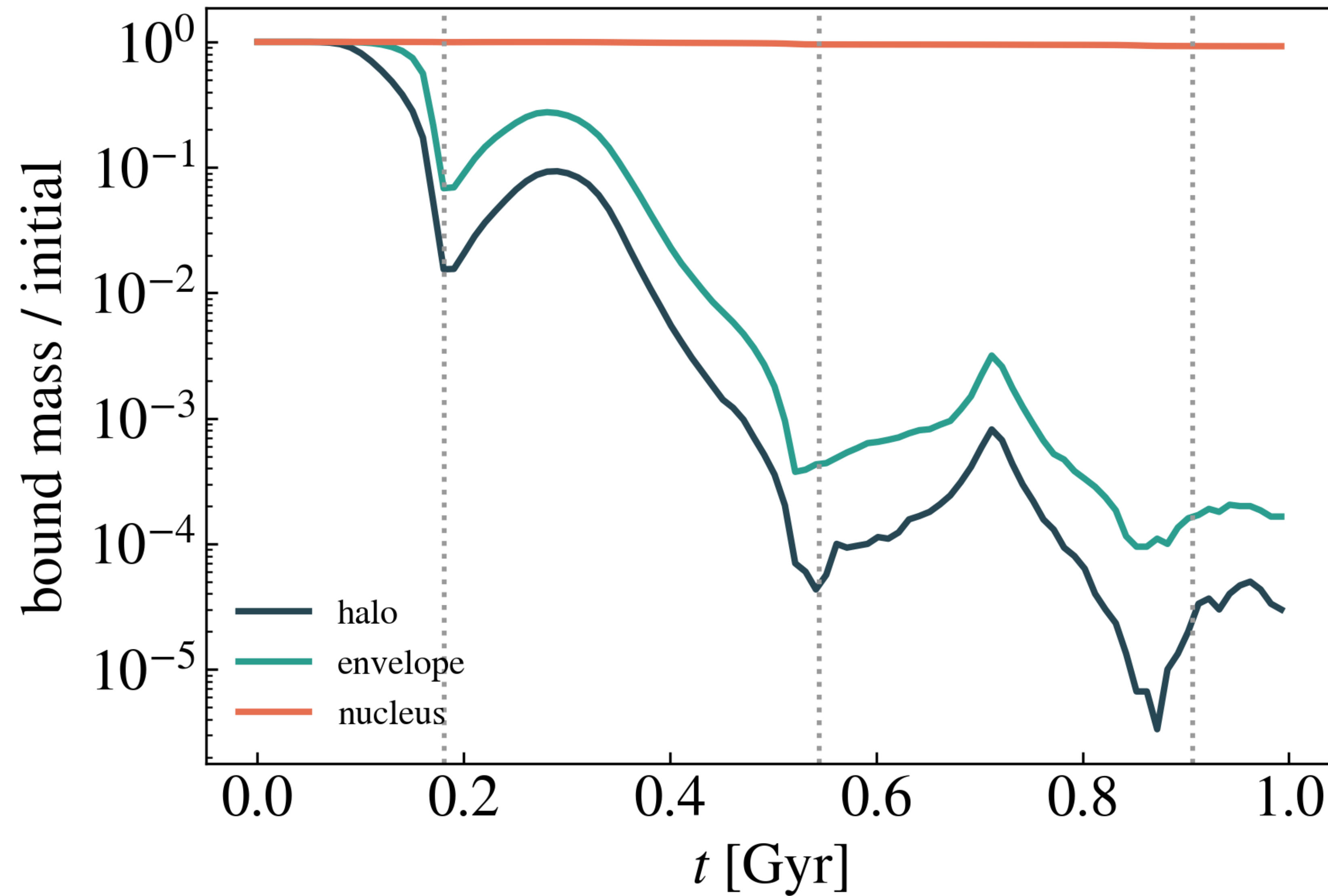
- Use AGAMA to see what a “reasonable” amount of residual DM is (*for now stay close to Peñurruvia et al. and Bekki & Freeman*)
- Ingredients:
 1. Static Milky Way potential with DM halo, spherical bulge, and disk (*Bovy 2015*)
 2. Progenitor with a Plummer nucleus, a King stellar envelope, and a DM halo

Tidal stripping with AGAMA



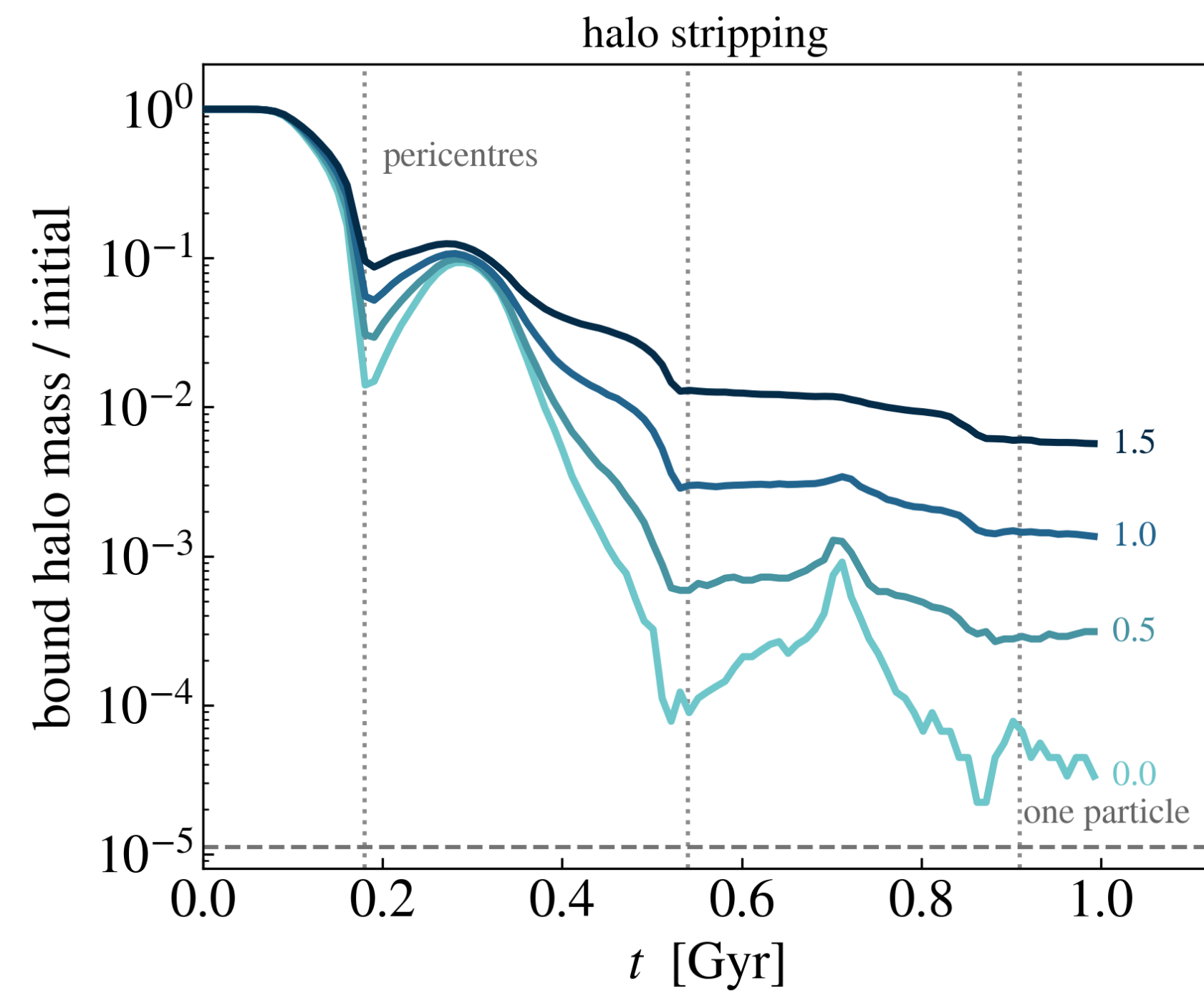
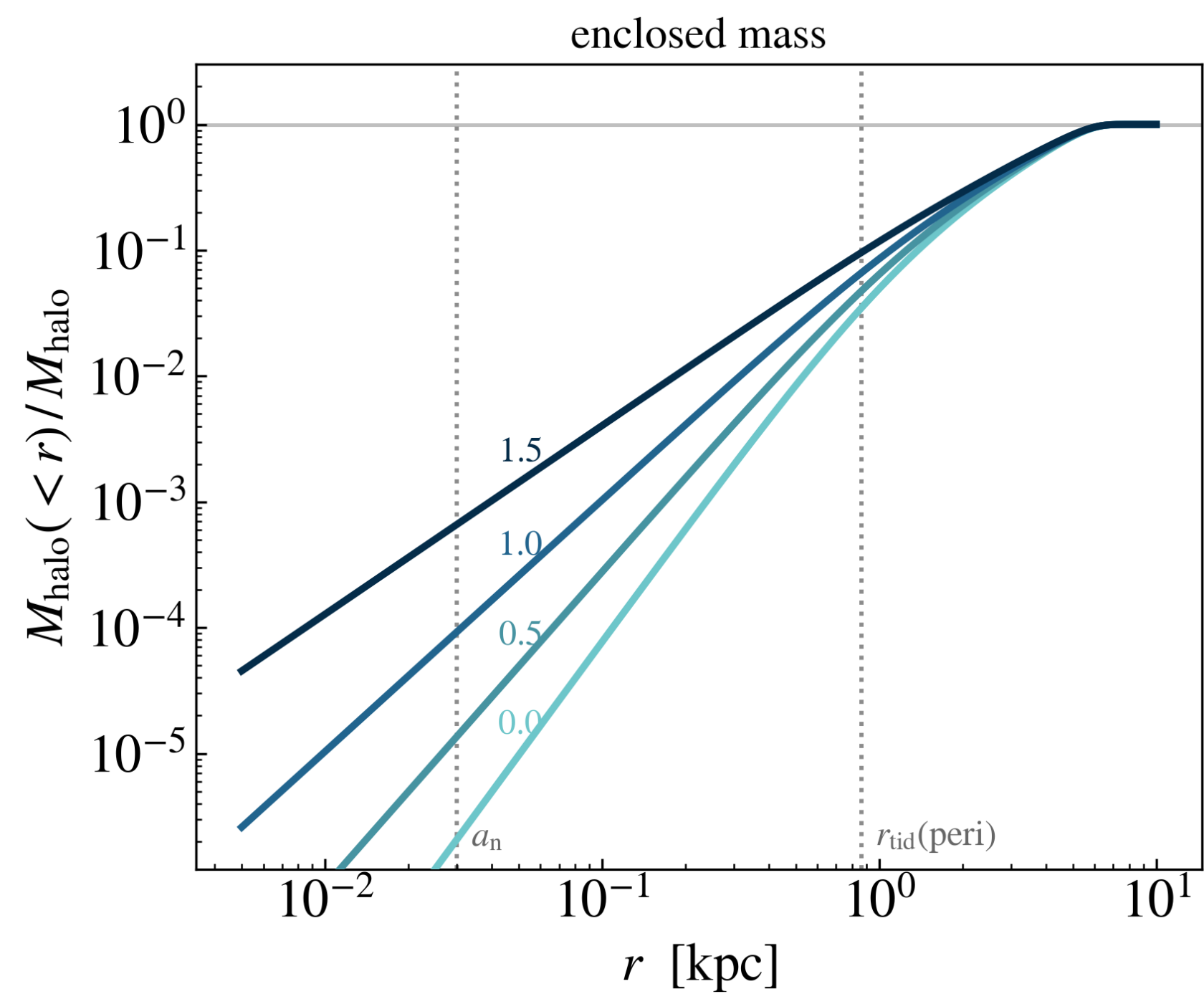
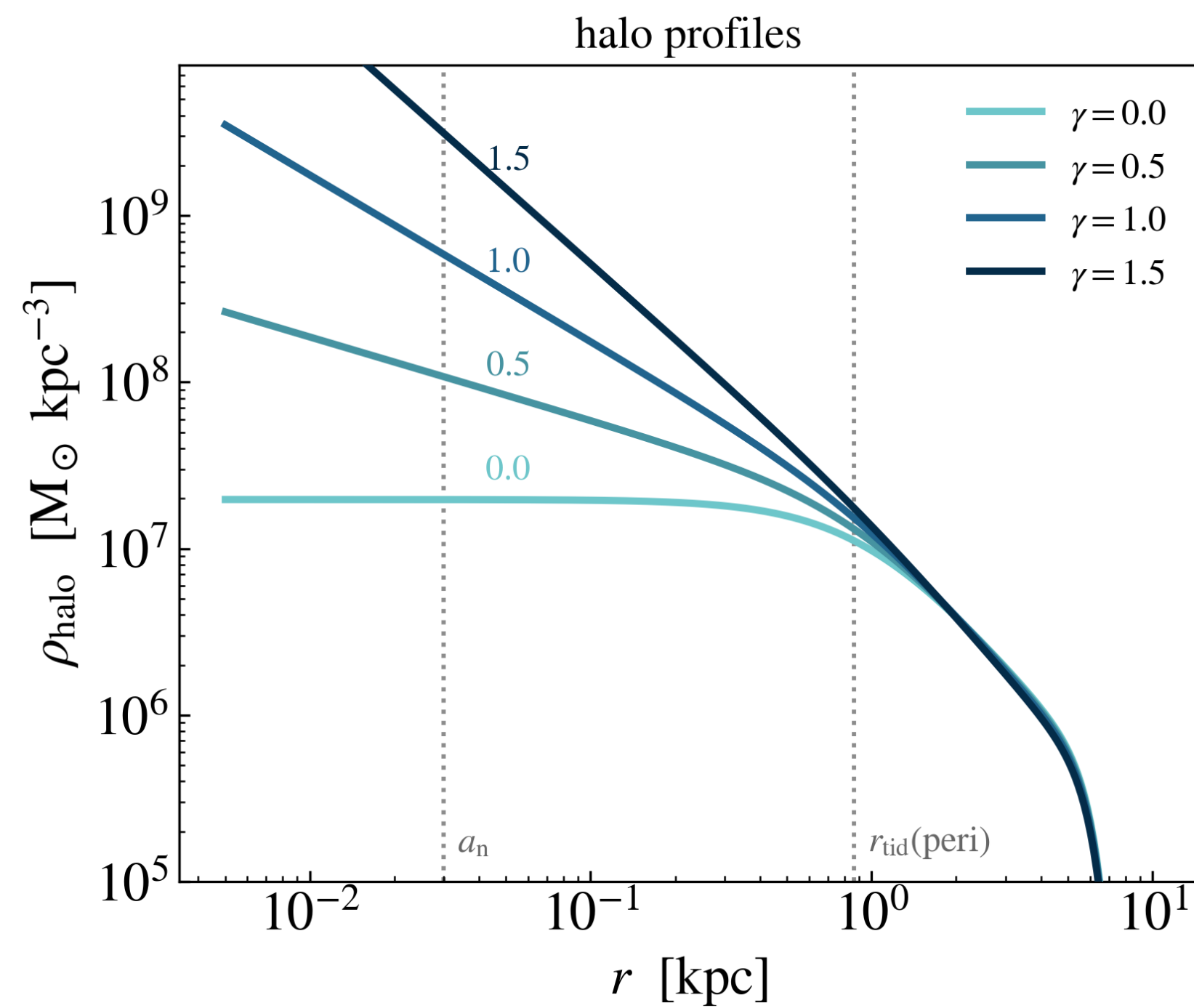
Coloured = bound, grey = unbound

Tidal stripping with AGAMA

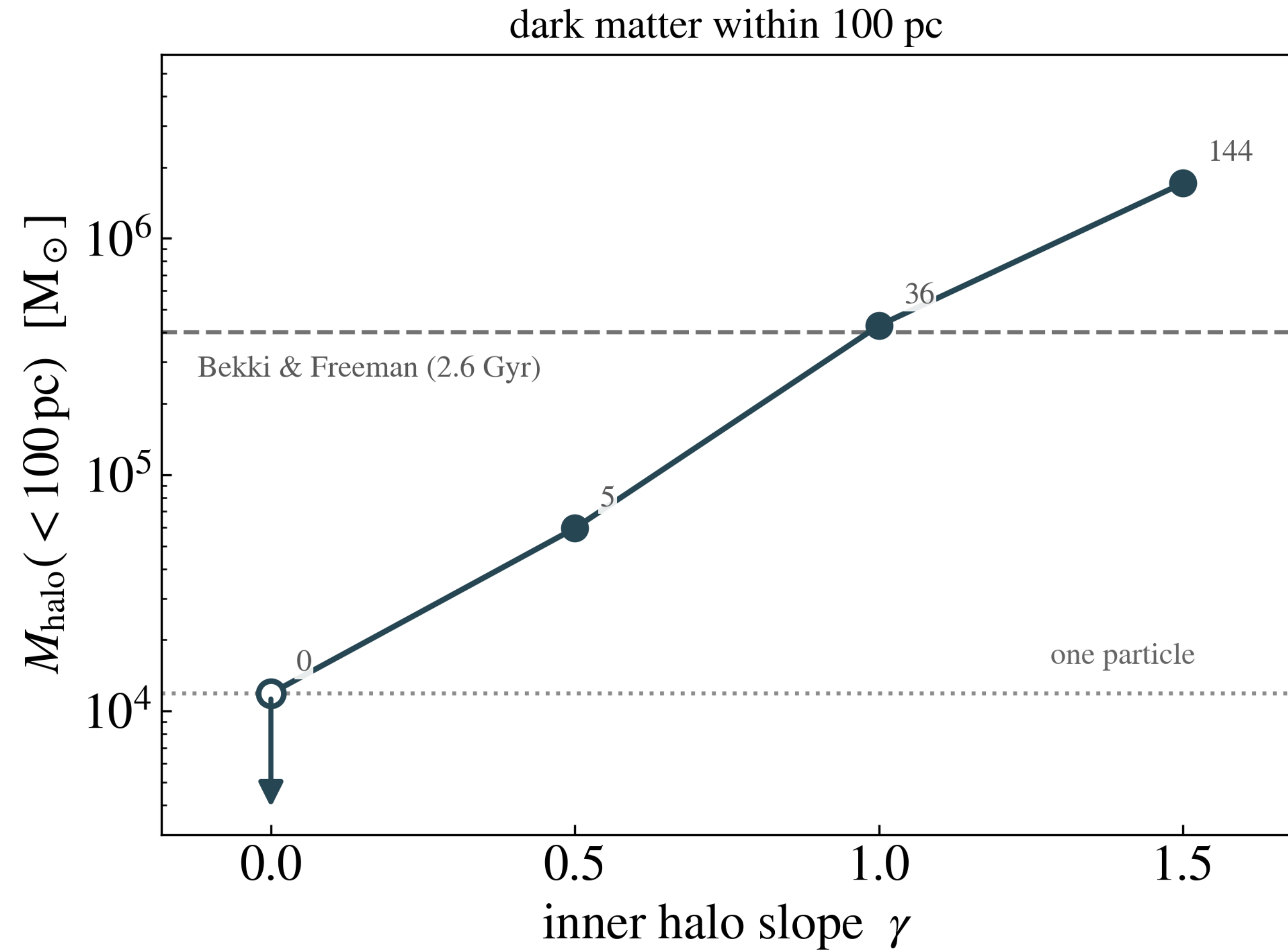
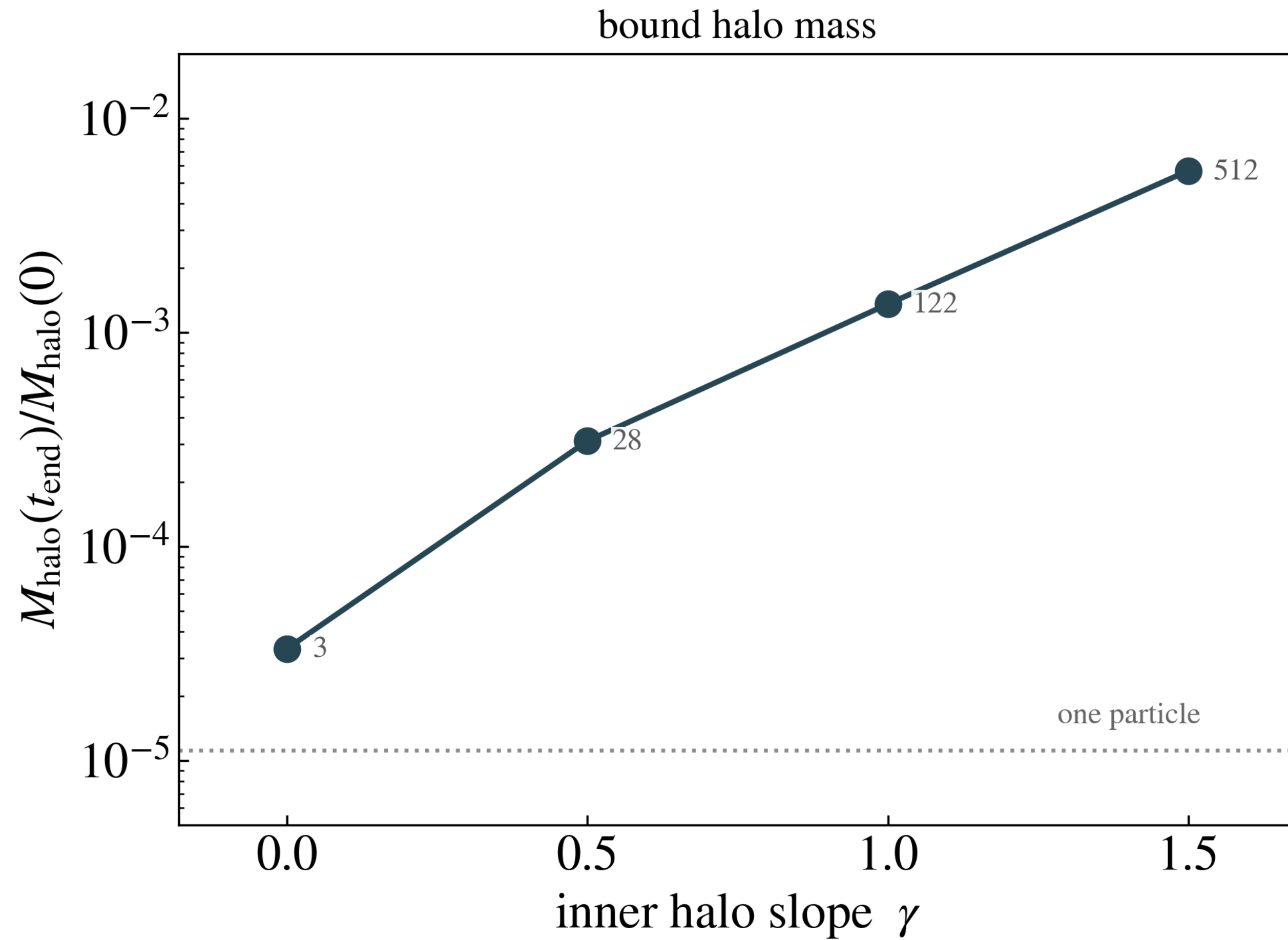


Dotted lines =
Pericentres

Tidal stripping with AGAMA



Tidal stripping with AGAMA



Backup

Set up

Static Milky Way

Progenitor dwarf

$$\text{Stellar mass } M_{\star} = \frac{M_{\omega}}{(1 - f_{\text{lost}})f_n}$$
$$\text{s.t. } M_{\text{nuc}} = f_n M_{\star} \text{ and } M_{\text{env}} = (1 - f_n) M_{\star}$$

$$f_n = 0.05,$$
$$f_{\text{lost}} = 0.2$$

$$\text{Halo with } M_{\text{halo}} = (f_d - 1) M_{\text{env}}$$

$$f_d = 10$$

agama.SelfConsistentModel

Evolution

- Use kick-drift-kick leapfrog (for timestep Δt):

$$1. \quad \mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{\Delta t}{2} \mathbf{a}_n$$

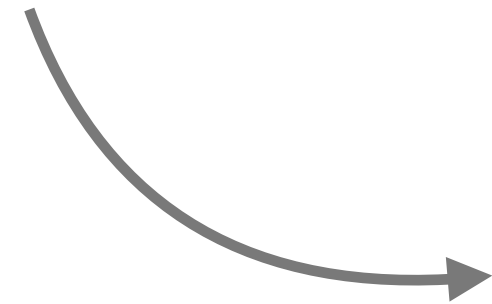
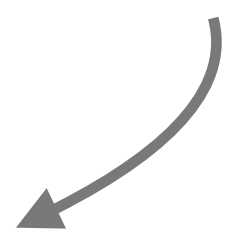
$$2. \quad \mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t \mathbf{v}_{n+1/2}$$

$$3. \quad \mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{\Delta t}{2} \mathbf{a}_{n+1}$$

where $\mathbf{a}_i = \mathbf{a}_{\text{self}}(\mathbf{x}_i) + \mathbf{a}_{\text{ext}}(\mathbf{x}_i)$

pyfalcon

static MW



What we measure

- Center position of dwarf remnant (from nucleus particles by shrinking sphere approach)
- Bound mass (compute potential from currently bound set, drop positive energy particles in dwarf frame, repeat three times)
- At final snapshot, measure mass within 100 pc about remnant centre

Parameters

Quantity	Symbol / setting	Fiducial
Nucleus mass, scale radius	M_{nuc}, a_n	$6.25 \times 10^6 M_{\odot}, 30 \text{ pc}$
Envelope mass, scale length	$M_{\text{env}}, a_{\text{env}}$	$1.19 \times 10^8 M_{\odot}, 770 \text{ pc}$
Halo mass	M_{halo}	$1.069 \times 10^9 M_{\odot}$
Halo core, truncation	r_c, r_{trunc}	0.99 kpc, 6.16 kpc
Halo inner slope	γ	0 (cored)
Particle counts	halo / envelope / nucleus	30 000 / 20 000 / 20 000
Softening	ϵ per component	0.10 / 0.05 / 0.010 kpc
Time step	$\tau = 2^{-13}$	0.119 Myr
Record interval	84 steps	10.0 Myr
Start radius, speed, tilt	$r_{\text{in}}, v_{\text{in}}, \theta$	26.25 kpc, 60 km s^{-1} , 30°
Run lengths	infall / isolation	1.0 / 0.5 Gyr

Stopping
“slingshot” effect
from particles
being artificially
heavy

Testing stability of
progenitor

Tidal stripping with AGAMA - disk

